

TOP 30

DevOps

INTERVIEW QUESTION



Created by- **TOPPERWORLD**

Q 1. Who is a DevOps engineer?

Ans : A DevOps engineer is a person who works with both software developers and the IT staff to ensure smooth code releases.

They are generally developers who develop an interest in the deployment and operations domain or the system admins who develop a passion for coding to move towards the development side.

In short, a DevOps engineer is someone who has an understanding of SDLC (Software Development Lifecycle) and of automation tools for developing CI/CD pipelines.

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Q 2. Why DevOps has become famous?

Ans : These days, the market window of products has reduced drastically. We see new products almost daily. This provides a myriad of choices to consumers but it comes at a cost of heavy competition in the market. Organizations can not afford to release big features after a gap.

They tend to ship off small features as releases to the customers at regular intervals so that their products don't get lost in this sea of competition.

Customer satisfaction is now a motto to the organizations which has also become the goal of any product for its success. In order to achieve this, companies need to do the below things:

- Frequent feature deployments
- Reduce time between bug fixes
- Reduce failure rate of releases
- Quicker recovery time in case of release failures.

In order to achieve the above points and thereby achieving seamless product delivery, DevOps culture acts as a very useful tool.

Due to these advantages, multi-national companies like Amazon and Google have adopted the methodology which has resulted in their increased performance.

Q 3. What is the use of SSH?

Ans : SSH stands for Secure Shell and is an administrative protocol that lets users have access and control the remote servers over the Internet to work using the command line.

SSH is a secured encrypted version of the previously known Telnet which was unencrypted and not secure. This ensured that the communication with the remote server occurs in an encrypted form.

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SSH also has a mechanism for remote user authentication, input communication between the client and the host, and sending the output back to the client.

Q 4. What is configuration management?



Ans : Configuration management (CM) is basically a practice of systematic handling of the changes in such a way that system does not lose its integrity over a period of time.

This involves certain policies, techniques, procedures, and tools for evaluating change proposals, managing them, and tracking their progress along with maintaining appropriate documentation for the same.

CM helps in providing administrative and technical directions to the design and development of the appreciation.

The following diagram gives a brief idea about what CM is all about:



Q 5. What is the importance of having configuration management in DevOps?



Ans : Configuration management (CM) helps the team in the automation of time-consuming and tedious tasks thereby enhancing the organization's performance and agility.

It also helps in bringing consistency and improving the product development process by employing means of design streamlining, extensive documentation, control, and change implementation during various phases/releases of the project.

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Q 6. What does CAMS stand for in DevOps?

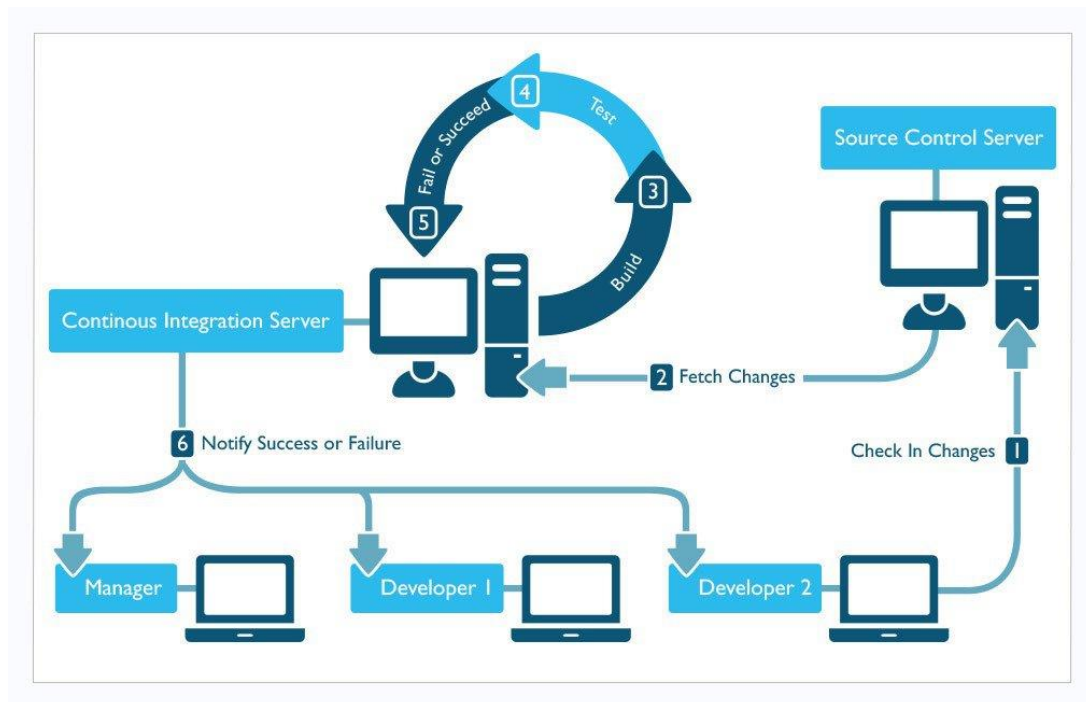
Ans : CAMS stands for Culture, Automation, Measurement, and Sharing. It represents the core deeds of DevOps.

Q 7. What is Continuous Integration (CI)?

Ans : Continuous Integration (CI) is a software development practice that makes sure developers integrate their code into a shared repository as and when they are done working on the feature.

Each integration is verified by means of an automated build process that allows teams to detect problems in their code at a very early stage rather than finding them after the deployment.





Based on the above flow, we can have a brief overview of the CI process.

- Developers regularly check out code into their local workspaces and work on the features assigned to them.
- Once they are done working on it, the code is committed and pushed to the remote shared repository which is handled by making use of effective version control tools like git.
- The CI server keeps track of the changes done to the shared repository and it pulls the changes as soon as it detects them.
- The CI server then triggers the build of the code and runs unit and integration test cases if set up.
- The team is informed of the build results. In case of the build failure, the team has to work on fixing the issue as early as possible, and then the process repeats.



Q 8. Why is Continuous Integration needed?

Ans : By incorporating Continuous Integration for both development and testing, it has been found that the software quality has improved and the time taken for delivering the features of the software has drastically reduced.

This also allows the development team to detect and fix errors at the initial stage as each and every commit to the shared repository is built automatically and run against the unit and integration test cases.

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Q 9. What is Continuous Testing (CT)?

Ans : Continuous Testing (CT) is that phase of DevOps which involves the process of running the automated test cases as part of an automated software delivery pipeline with the sole aim of getting immediate feedback regarding the quality and validation of business risks associated with the automated build of code developed by the developers.

Using this phase will help the team to test each build continuously (as soon as the code developed is pushed) thereby giving the dev teams a chance to get instant feedback on their work and ensuring that these problems don't arrive in the later stages of SDLC cycle.

Doing this would drastically speed up the workflow followed by the developer to develop the project due to the lack of manual intervention steps to rebuild the project and run the automated test cases every time the changes are made.

Q 10. What are the three important DevOps KPIs?

Ans : Few KPIs of DevOps are given below:

- Reduce the average time taken to recover from a failure.
- Increase Deployment frequency in which the deployment occurs.
- Reduced Percentage of failed deployments.

Q 11. Explain the different phases in DevOps methodology.

Ans : DevOps mainly has 6 phases and they are:

➤ **Planning:**

- This is the first phase of a DevOps lifecycle that involves a thorough understanding of the project to ultimately develop the best product.
- When done properly, this phase gives various inputs required for the development and operations phases.
- This phase also helps the organization to gain clarity regarding the project development and management process.
- Tools like Google Apps, Asana, Microsoft teams, etc are used for this purpose.

➤ **Development:**

- The planning phase is followed by the Development phase where the project is built by developing system infrastructure, developing features by writing codes, and then defining test cases and the automation process.
- Developers store their codes in a code manager called remote repository which aids in team collaboration by allowing view, modification, and versioning of the code.
- Tools like git, IDEs like the eclipse, IntelliJ, and technological stacks like Node, Java, etc are used.

➤ **Continuous Integration (CI):**

- This phase allows for automation of code validation, build, and testing. This ensures that the changes are made properly without development environment errors and also allows the identification of errors at an initial stage.
- Tools like Jenkins, circleCI, etc are used here.

➤ Deployment:

- DevOps aids in the deployment automation process by making use of tools and scripts which has the final goal of automating the process by means of feature activation.
- Here, cloud services can be used as a force that assists in upgrade from finite infrastructure management to cost-optimized management with the potential to infinite resources.
- Tools like Microsoft Azure, Amazon Web Services, Heroku, etc are used.

➤ Operations:

- This phase usually occurs throughout the lifecycle of the product/software due to the dynamic infrastructural changes.
- This provides the team with opportunities for increasing the availability, scalability, and effective transformation of the product.
- Tools like Loggly, BlueJeans, Appdynamics, etc are used commonly in this phase.

➤ Monitoring:

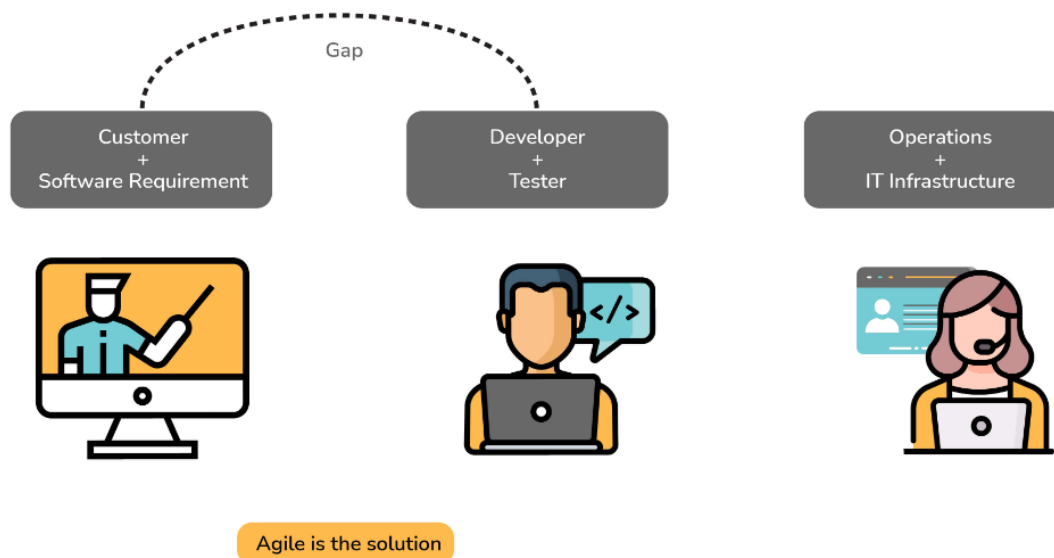
- Monitoring is a permanent phase of DevOps methodology. This phase is used for monitoring and analyzing information to know the status of software applications.
- Tools like Nagios, Splunk, etc are commonly used.

Q 12. How is DevOps different than the Agile Methodology?

Ans : DevOps is a practice or a culture that allows the collaboration of the development team and the operations team to come together for successful product development. This involves making use of practices like continuous development, integration, testing, deployment, and monitoring of the SDLC cycle.

DevOps tries to reduce the gap between the developers and the operations team for the effective launch of the product.

Agile is nothing but a software development methodology that focuses on incremental, iterative, and rapid releases of software features by involving the customer by means of feedback. This methodology removes the gap between the requirement understanding of the clients and the developer

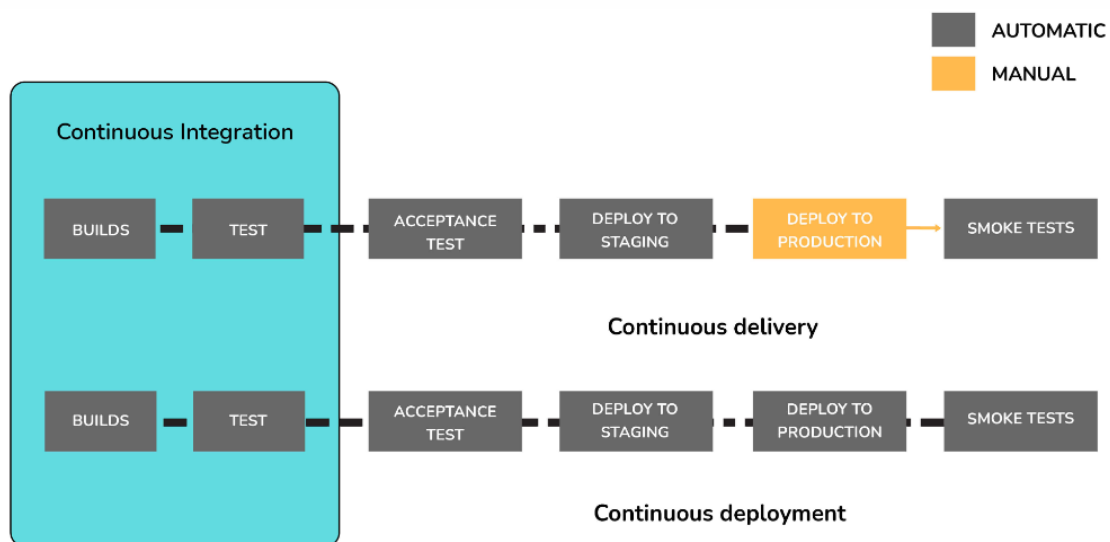


Q13. Differentiate between Continuous Deployment and Continuous Delivery?

Ans : The main difference between Continuous Deployment and Continuous Delivery are given below:

Continuous Deployment	Continuous Delivery
The deployment to the production environment is fully automated and does not require manual/ human intervention.	In this process, some amount of manual intervention with the manager's approval is needed for deployment to a production environment.

Continuous Deployment	Continuous Delivery
Here, the application is run by following the automated set of instructions, and no approvals are needed.	Here, the working of the application depends on the decision of the team.



Q 14. What can you say about antipatterns of DevOps?

Ans : A pattern is something that is most commonly followed by large masses of entities. If a pattern is adopted by an organization just because it is being followed by others without gauging the requirements of the organization, then it becomes an anti-pattern.

Similarly, there are multiple myths surrounding DevOps which can contribute to antipatterns, they are:

- DevOps is a process and not a culture.
- DevOps is nothing but Agile.
- There should be a separate DevOps group.
- DevOps solves every problem.
- DevOps equates to developers running a production environment.



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